

- 6 Aquaculture is the controlled cultivation of aquatic organisms such as fish, shellfish and plants. It is an important source of food in many countries of the world.

The photograph shows freshwater aquaculture ponds in Africa.



(Source: © E.Westmacott/Alamy Stock Photo)

Vibrio are bacteria that are common in aquatic habitats. They cause diarrhoea and cholera and are also pathogenic to shellfish.

The resistance of *Vibrio* to antibiotics and to heavy metals in aquaculture ponds in Nigeria was investigated.

- (a) Samples of water were taken from 120 ponds. Of these, 74 of the ponds had been treated previously with antibiotics.

Which of the following is the percentage of **untreated** ponds?

(1)

- ☐ A 38
- ☐ B 61
- ☐ C 62
- ☐ D 163

- (b) *Vibrio* were grown on agar containing different heavy metals at a range of concentrations.

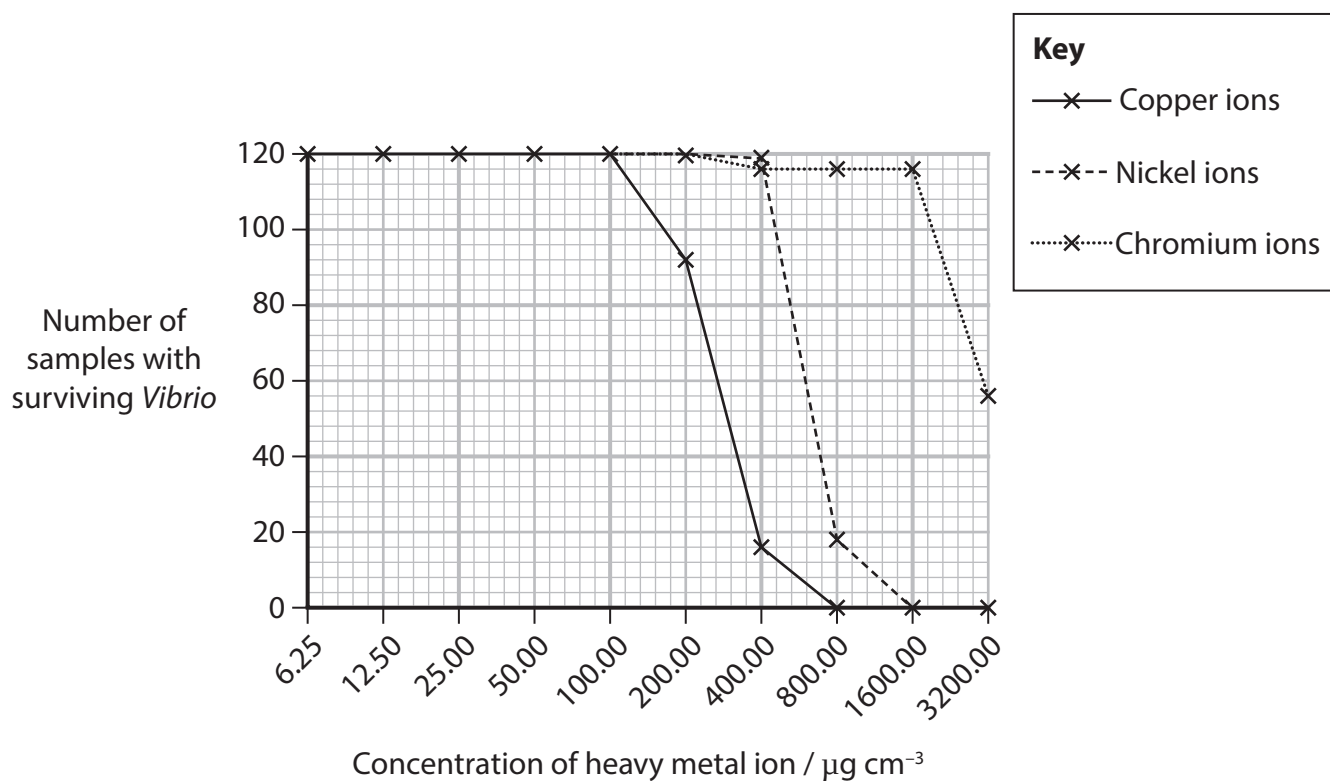
The highest concentration of heavy metal tested was 3.2 mg cm^{-3} and the lowest concentration was $6.25 \text{ } \mu\text{g cm}^{-3}$.

- (i) How many times does the highest concentration need to be diluted to make the lowest concentration?

(1)

- ☐ A 5.12
- ☐ B 51.2
- ☐ C 512
- ☐ D 5120

- (ii) The graph shows the effect of different concentrations of heavy metals on the survival of the *Vibrio* collected from each of the 121 ponds.



Describe the conclusions that can be made about the effects of heavy metal ions on the survival of *Vibrio*.

(3)

[illegible]

(c) The samples of *Vibrio* taken from each pond were spread on agar containing different antibiotics.

- (i) The scientists expected the percentage of resistant bacteria to be higher in the samples of *Vibrio* taken from the ponds previously treated with antibiotics.

Explain why the scientists expected this difference.

(3)

- (ii) Some of the antibiotics used were bacteriostatic and some were bactericidal.

State the difference between a bacteriostatic antibiotic and a bactericidal antibiotic.

(1)

- (iii) The method did not distinguish between bacteriostatic and bactericidal antibiotics because no colonies grew.

Suggest how the method would need to be modified to show which antibiotics were bacteriostatic and which were bactericidal.

Give a reason for your answer.

(2)

(Total for Question 6 = 11 marks)